

A MACHINE LEARNING APPROACH USING STATISTICAL MODELS FOR EARLY DETECTION OF CARDIAC ARREST IN NEWBORN BABIES IN THE CARDIAC INTENSIVE CARE UNIT

ABSTRACT

Cardiac arrest in newborn babies is an alarming yet typical medical emergency. Early detection is critical for providing these babies with the best care and treatment. Recent research has focused on identifying the potential indicators and biomarkers of cardiac arrest in newborn babies and developing accurate and efficient diagnostic tools for early detection. An array of imaging techniques, such as echocardiography and computed tomography may help provide early detection of cardiac arrest. This research aims to develop a Cardiac Machine Learning model (CMLM) using statistical models for the early detection of cardiac arrest in newborn babies in the Cardiac Intensive Care Unit (CICU). The cardiac arrest events were identified using a combination of the neonate's physiological parameters. Statistical modeling techniques, such as logistic regression and support vector machines, were used to construct predictive models for cardiac arrest. The proposed model will be used in the CICU to enable early detection of cardiac arrest in newborn babies. In a training (Tr) comparison region, the proposed CMLA reached 0.912 delta-p value, 0.894 False discovery rate (FDR) value, 0.076 False omission rate (FOR) value, 0.859 prevalence threshold value and 0.842 CSI value. In a testing (Ts) comparison region, the proposed CMLA reached 0.896 delta-p values, 0.878 FDR value, 0.061 FOR value, 0.844 prevalence threshold values and 0.827 CSI value. It will help reduce the mortality and morbidity of newborn babies due to cardiac arrest in the CICU.

SYSTEM ANALYISS

EXISTING SYSTEM

Carlisle et al. [21] has discussed the Heart failure is when the heart cannot pump enough blood to the rest of the body. Various conditions, including high blood pressure, coronary artery disease, and diabetes, can cause it. Atrial fibrillation is an arrhythmia (irregular heartbeat) in which the heart's upper chambers (atria) beat rapidly and irregularly. It can cause a decrease in the amount of blood pumped to the rest of the body, leading to symptoms such as shortness of breath and fatigue. Atrial fibrillation is a common cause of heart failure. Heart failure and atrial fibrillation treatment usually involve medications to control the heart rate and rhythm, lifestyle changes, and sometimes surgery to repair or replace the heart.

Yaku et al. [22] has discussed the Risk factors for functional decline during hospitalization in very old patients with acute decompensated heart failure include age, gender, co-morbidities, and frailty. In addition, complex medical problems, the need for aggressive treatments, and the presence of cognitive impairment may increase the risk of functional decline. Clinical outcomes associated with a functional decline during hospitalization in very old patients with acute decompensated heart failure include increased length of stay, healthcare utilization, mortality, and decreased quality of life. The functional decline may also lead to higher rates of re-hospitalization, as well as an increased risk of institutionalization. Additionally, the functional decline may lead to an increased risk of falls and delirium due to decreased mobility and activity levels.

Fonarow et al. [23] has discussed the Risk stratification for in-hospital mortality in acutely decompensated heart failure determines which patients are at higher risk of dying while in the hospital. It is done by using classification and regression tree analysis. Classification and regression tree analysis is a type of predictive analytics that uses trees to classify and predict outcomes. The trees are nodes representing various conditions, characteristics, or features associated with the outcome. Using a combination of these nodes, the model can determine the likelihood of a particular outcome occurring. The model can then be used to identify patients at a higher risk of in-hospital mortality and to guide the treatment of the patient.

Gaies et al. [24] has discussed the Vasoactive–inotropic score (VIS) is designed to predict morbidity and mortality in infants after cardiopulmonary bypass (CPB). The VIS is calculated from the levels of vasoactive and inotropic drugs administered to the infant during and after

CPB. These drugs are used to regulate the patient's blood pressure and heart rate. The VIS is believed to accurately predict post-CPB morbidity and mortality because it reflects the degree of hemodynamic instability in the infant. Higher VIS scores indicate greater hemodynamic instability and, therefore, a greater risk of morbidity and mortality. Studies have found that higher VIS scores are associated with increased mortality, extended hospital stays, and an increased need for vasopressor and inotropic support. The VIS is a significant predictor of outcome after CPB and can help clinicians identify infants who may require closer monitoring and more aggressive management.

Shah et al. [25] has discussed the Phenomapping is a novel classification system for heart failure with preserved ejection fraction (HFpEF). It is based on the analysis of phenotypic characteristics, such as demographics, clinical profile, laboratory values, electrocardiographic findings, echocardiography findings, and biomarkers. The goal of Phenomapping is to provide a more comprehensive and meaningful classification system for HFpEF that is based on the distinct phenotypes of the disease. This classification system will enable clinicians to more accurately diagnose and stratify patients with HFpEF, leading to better management and improved outcomes. The Phenomapping system also provides a platform for further research into the underlying path physiology of HFpEF, allowing for a better understanding of the disease and the potential for improved treatments.

Disadvantages

- The complexity of data: Most of the existing machine learning models must be able to accurately interpret large and complex datasets to detect cardiac arrest in newborn babies.
- Data availability: Most machine learning models require large amounts of data to create accurate predictions. If data is unavailable in sufficient quantities, then model accuracy may suffer.
- Incorrect labeling: The existing machine learning models are only as accurate as the data trained using the input dataset. If the data has been incorrectly labeled, the model cannot make accurate predictions.

PROPOSED SYSTEM

In the existing system, ml models can be used to identify the most critical risk factors associated with the condition and predict the likelihood of an infant experiencing an arrest. Therefore,

these statistical models should be used to improve newborns' early detection and intervention of cardiac arrest [17].

Machine learning is increasingly used to predict and detect cardiac arrest in newborn babies. Cardiac arrest is a life-threatening condition in which the heart suddenly stops beating, and blood flow to the brain and other organs stops. It can lead to permanent brain damage or death. Due to the complexity of the condition, early detection of cardiac arrest in newborns has been difficult. However, machine learning is changing that [18].

Machine learning algorithms analyze large amounts of complex data, such as patient medical histories, vital signs, and other physiological data. The algorithms can detect patterns in the data indicative of cardiac arrest and alert medical personnel. For example, one study used machine learning to detect signs of cardiac arrest in newborns by analyzing their heart rates, breathing patterns, and other vital signs. The algorithm detected signs of cardiac arrest up to eight hours before conventional methods. It could significantly improve the chances of survival for newborns and reduce the damage caused by the condition. In addition, machine learning is used to predict newborns' risk of cardiac arrest. By analyzing large amounts of patient data, machine learning algorithms can identify risk factors associated with the condition. It can help medical personnel identify newborns at an increased risk of cardiac arrest to receive the care they need. The machine learning is revolutionizing the early detection of cardiac arrest in newborns [19].

By analyzing large amounts of complex data, machine learning algorithms can detect signs of cardiac arrest and identify newborns at an increased risk of the condition. This technology could save lives and reduce the damage caused by cardiac arrest in newborns. The critical contribution of machine learning models used for the Early Detection of Cardiac Arrest in Newborn Babies is that these models can detect subtle changes in vital signs such as heart rate, respiratory rate, and oxygen saturation that are difficult to detect with the naked eye. This early detection can help to identify newborns at risk of cardiac arrest and allow for timely intervention and treatment [20]. Additionally, machine learning models can be used to analyze patient data to provide personalized advice and care to patients, enabling better long term management of their condition.

Advantages

- Automated and accurately detected critical signs associated with cardiac arrest in newborn babies.
- Ability to recognize subtle changes in the baby's vital signs that can indicate potential cardiac arrest.
- Ability to identify high-risk babies likely to suffer from cardiac arrest.
- Early detection of cardiac arrest, enabling timely interventions that can improve the outcome.
- Reduction in the time and cost associated with traditional monitoring methods.
- Improved patient outcomes due to early diagnosis and treatment of cardiac arrest.

SYSTEM REQUIREMENTS

➤ H/W System Configuration:-

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|-------------|-----------------------------|
| ➤ Processor | - Pentium –IV |
| ➤ RAM | - 4 GB (min) |
| ➤ Hard Disk | - 20 GB |
| ➤ Key Board | - Standard Windows Keyboard |
| ➤ Mouse | - Two or Three Button Mouse |
| ➤ Monitor | - SVGA |

SOFTWARE REQUIREMENTS:

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|--------------------|--------------------------|
| ❖ Operating system | : Windows 7 Ultimate. |
| ❖ Coding Language | : Python. |
| ❖ Front-End | : Python. |
| ❖ Back-End | : Django-ORM |
| ❖ Designing | : Html, css, javascript. |
| ❖ Data Base | : MySQL (WAMP Server). |

Under the guidance of
Mr. P ABDUL SUBHAHANALLA
ASSOCIATE PROFESSOR
DEPT . OF CSE

Submitted by:
BURRA HARSHITHA (21XU1AO518)
KATARAM DIKSHITHA (21XU1AO536)
GUDAVALLI KALYAN (21XU1AO543)